DEPARTMENT OF COMPUTER APPLICATIONS

PROJECT SYNOPSIS

QueueLess

Smart Salon Appointment & Queue Management System

Submitted in partial fulfilment of the requirements for
the award of the degree of
Master of Computer Applications (MCA)

Submitted by

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1. Project Overview

Problem Statement

Traditional salon appointment and queue management is largely handled through phone calls, WhatsApp messages, or manual paper registers. This informal approach leaves customers without visibility into their exact turn or estimated waiting time, often forcing them to wait in person for long, unpredictable durations. At the same time, salon owners struggle to manage appointments, walk-in customers, staff availability, cancellations, and overlapping bookings, which leads to scheduling conflicts, idle staff time, and a poor overall customer experience.

Proposed Solution

QueueLess — Smart Salon Appointment & Queue Management System is a web-based platform that digitizes and automates the entire salon booking and queue workflow. Customers can browse nearby salons, view real-time service availability, book appointments online, and track their live queue position and estimated wait time from their phone or computer — removing the need to wait physically inside the salon. Salon owners and staff, in turn, get a centralized dashboard to manage appointments, walk-ins, staff schedules, and service timings, reducing manual effort and scheduling errors.

Significance and Impact

By replacing manual, phone-based coordination with a structured digital system, QueueLess reduces customer waiting anxiety, minimizes no-shows and double-bookings, and helps salon owners make data-driven staffing and scheduling decisions. The system is designed to be lightweight and affordable, making it accessible to small and medium-sized salons that cannot invest in expensive enterprise software, while still offering a modern, app-like booking experience to customers.

> **In short:** QueueLess turns an unstructured, manual queueing process into a transparent, real-time, and data-driven digital experience for both customers and salon owners.

2. Project Objectives

The QueueLess project is guided by the following specific objectives:

1. **Enable online appointment booking** — Allow customers to search salons, view available services and staff, and book an appointment for a preferred date and time slot without a phone call.
2. **Provide real-time queue visibility** — Show customers their live position in the queue and an estimated waiting time, updated automatically as the queue progresses.
3. **Simplify salon-side management** — Give salon owners/admins a single dashboard to manage staff, services, working hours, appointments, and walk-in entries.
4. **Reduce scheduling conflicts** — Prevent overlapping bookings and double allocation of a staff member through automated slot validation.
5. **Support walk-in customers** — Allow the salon to add walk-in customers directly into the digital queue alongside pre-booked appointments.
6. **Improve communication** — Notify customers about booking confirmation, queue updates, and cancellations in-app/via email, reducing dependence on manual calls or messages.
7. **Provide actionable insights** — Offer salon owners basic analytics (peak hours, most-booked services, staff utilization) to support better business decisions.

Measuring Success

Objective	Success Indicator
Online booking	Customer can complete a booking in under 3 steps without staff assistance
Live queue tracking	Queue position & wait time visible and updated without page refresh
Admin dashboard	Owner can add/edit staff, services and view the day's schedule from one screen
Conflict prevention	Zero double-booking of the same staff member for overlapping slots

3. Project Scope

In Scope — Core Modules

Customer Module: Registration/login, salon & service browsing, appointment booking, live queue & wait-time tracking, booking history, cancel/reschedule.

Salon Admin Module: Salon profile setup, service & pricing management, staff management, working-hour configuration, appointment approval.

Queue Management Module: Auto-generated token/queue number, real-time queue status updates, walk-in insertion into queue.

Staff Module: Staff login, view assigned appointments, mark service as in-progress/completed.

Notification Module: Booking confirmation, queue-turn alerts, cancellation notices.

Reports & Analytics Module: Daily bookings summary, peak-hour trends, staff-wise service count (for admin).

Out of Scope (Phase 2 / Future Enhancements)

- Online payment gateway integration for advance booking payments.
- Native Android/iOS mobile applications (initial version is a responsive web app).
- Multi-language support and SMS-based notifications.
- AI-based personalized service recommendations.

4. Methodology

Development Approach

The project will follow the **Agile (Iterative-Incremental) methodology**, with development broken into short sprints, each delivering a working, testable increment of the system (e.g., authentication, then booking, then queue management). This approach allows continuous testing, early feedback, and flexibility to refine features such as the queue algorithm as the project progresses — better suited to a project of this scale than a rigid Waterfall model.

Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, JavaScript (Fetch API / DOM manipulation) for a responsive, mobile-friendly interface
Backend	Node.js with Express.js — RESTful API layer handling authentication, bookings, and queue logic
Database	MongoDB with Mongoose ODM — stores users, salons, staff, appointments, and queue records
Real-time Updates	Socket.IO — pushes live queue position/status changes to connected clients
Authentication	JSON Web Tokens (JWT) with bcrypt password hashing
Tools	VS Code, Postman (API testing), Git & GitHub (version control), MongoDB Compass

System Architecture



The system follows a 3-tier client–server architecture. The client sends HTTP requests to the Express.js REST API, which processes business logic (authentication, booking validation, queue updates) and interacts with MongoDB for persistent storage. Socket.IO maintains a live connection for instant queue-status broadcasts.

5. Hardware & Software Requirements

Hardware Requirements

Component	Specification
Development Machine	Intel i3/i5 or equivalent, 8 GB RAM, 256 GB SSD
Client Device	Any smartphone/laptop with a modern browser and internet access
Server (Deployment)	Cloud instance — 1–2 vCPU, 2 GB+ RAM (e.g., Render/Railway/AWS free-mid tier)

Software Requirements

Component	Specification
OS	Windows / Linux / macOS
Runtime	Node.js (v18+), npm
Database	MongoDB (local) / MongoDB Atlas (cloud)
Browser	Chrome, Edge, or Firefox (latest)
Editor & VCS	VS Code, Git & GitHub

6. Timeline

Phase	Wk 1-2	Wk 3-4	Wk 5-6	Wk 7-8	Wk 9-10	Wk 11	Wk 12
Requirement Analysis							
UI/DB Design							
Frontend Development							
Backend & API Dev.							
Integration & Queue Logic							
Testing & Debugging							
Deployment & Documentation							

7. Project Outcome

Upon completion, the project is expected to deliver the following outcomes:

- **Web Application Development:** A fully functional, deployed QueueLess web application demonstrating end-to-end appointment booking and real-time queue management for salons.
- **Reduced Customer Wait-Time Uncertainty:** Customers gain live visibility into their queue status, reducing physical waiting and improving satisfaction.
- **Operational Efficiency for Salons:** Salon owners gain a centralized tool to manage staff, appointments, and walk-ins, reducing scheduling conflicts and manual coordination effort.
- **Reusable Learning Outcome:** Hands-on experience in building a full-stack application (HTML/CSS/JS + Node.js/Express.js + MongoDB) with real-time features using Socket.IO.
- **Possible Research Paper Publication:** Documentation of the queue-optimization approach and system design as a basis for a potential research paper/report submission.

Conclusion

QueueLess addresses a genuine, everyday inefficiency in the local salon industry by replacing manual, phone-based appointment handling with a transparent, real-time digital system. Built entirely with accessible, industry-relevant web technologies, the project not only solves a practical problem for customers and salon owners but also serves as a strong demonstration of full-stack web development skills, covering frontend design, RESTful API development, database modelling, and real-time communication.